Copy Fail exploit detection in Microsoft Azure

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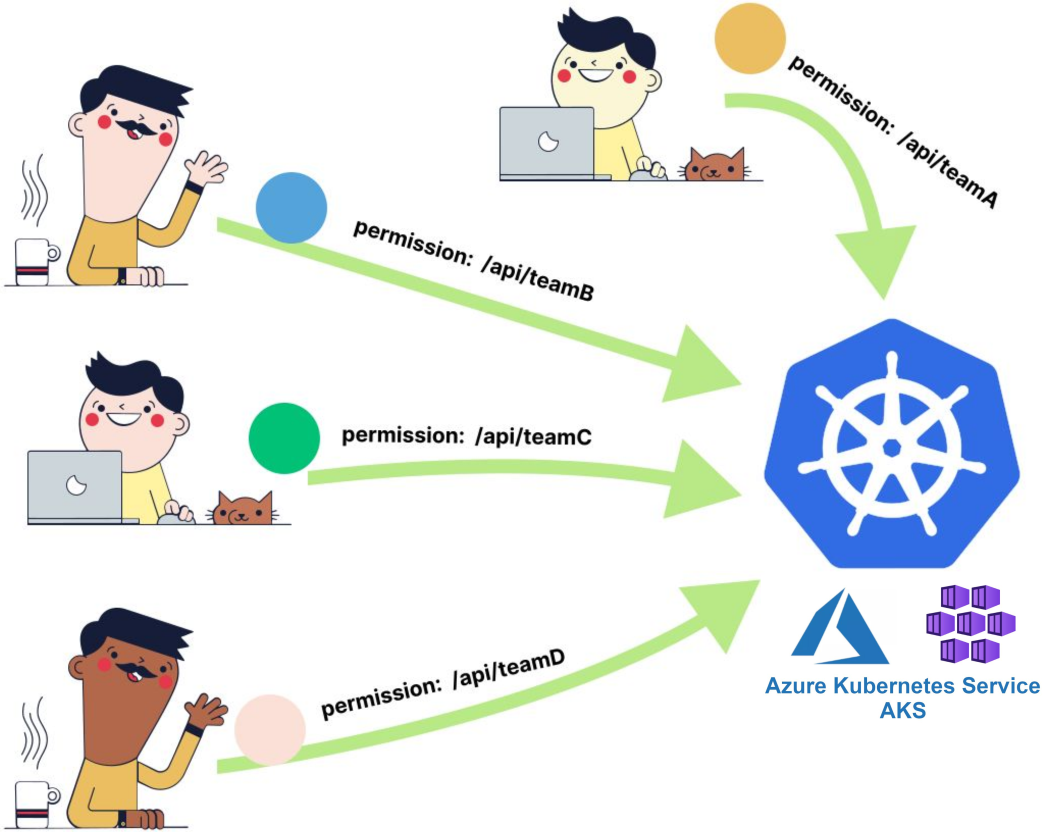
Copy Fail

The context



Multi tenant cluster
Employees can create unprivileged pods to test the applications they're working on.

What if one of them wants to get direct access to the underlying node?

Copy Fail

The **CVE-2026-31431** (Copy Fail) allows for a **100%** reliable local **privilege escalation** on **every Linux distribution** since **2017**.

CVSS Score: 7.8 (high)

Vulnerability: A bug in the authencesn cryptographic template enables an unprivileged local user to trigger a deterministic 4-byte write into the page cache.

Exploitation: With a 732-byte Python script, an attacker can modify a setuid binary to gain full root access.

Stealth: Because only the in-memory page cache is corrupted, standard file integrity tools miss the exploit as the on-disk file remains unchanged.

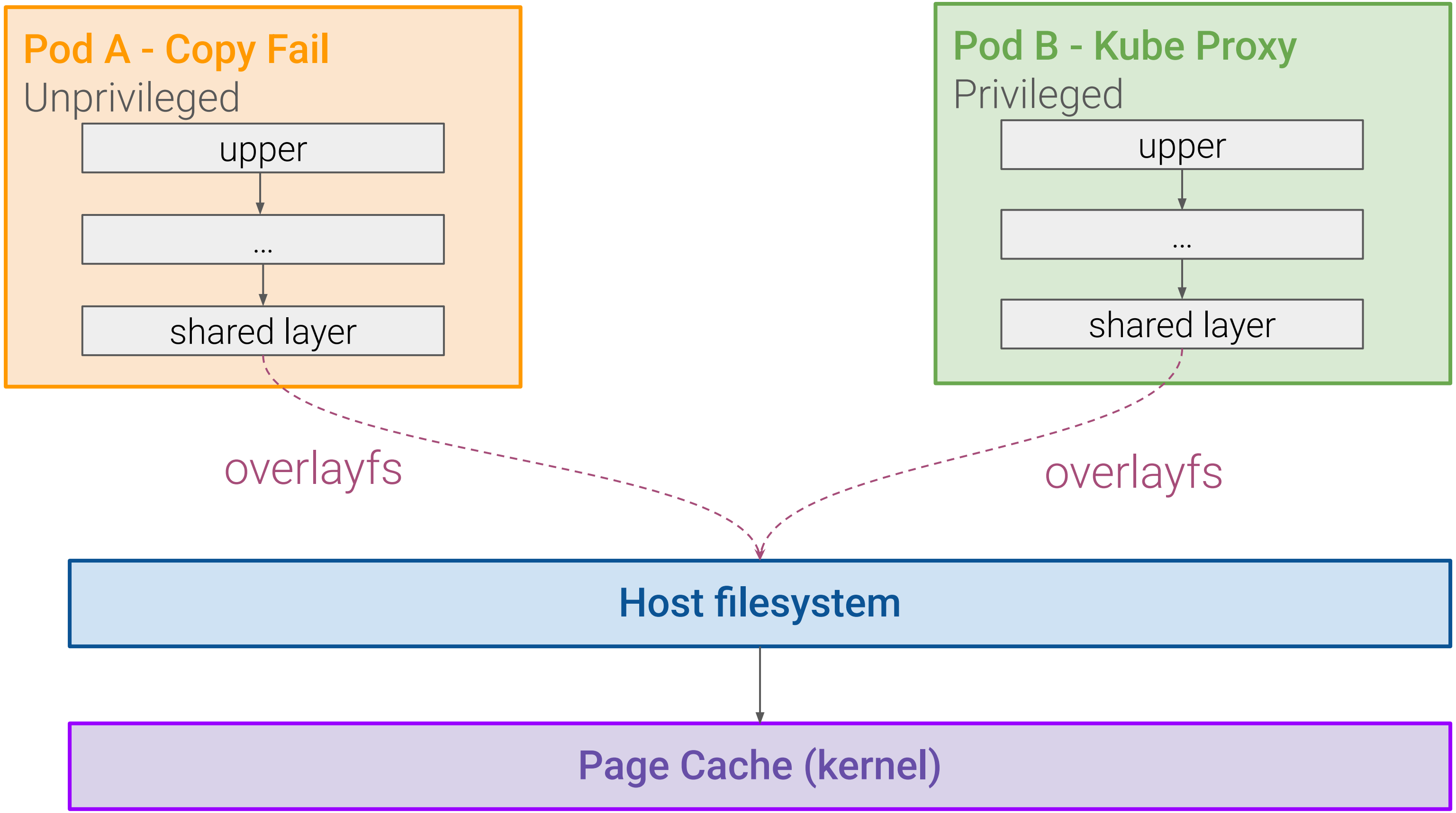
Copy Fail in Kubernetes

In K8s, the same vulnerability manifests differently!

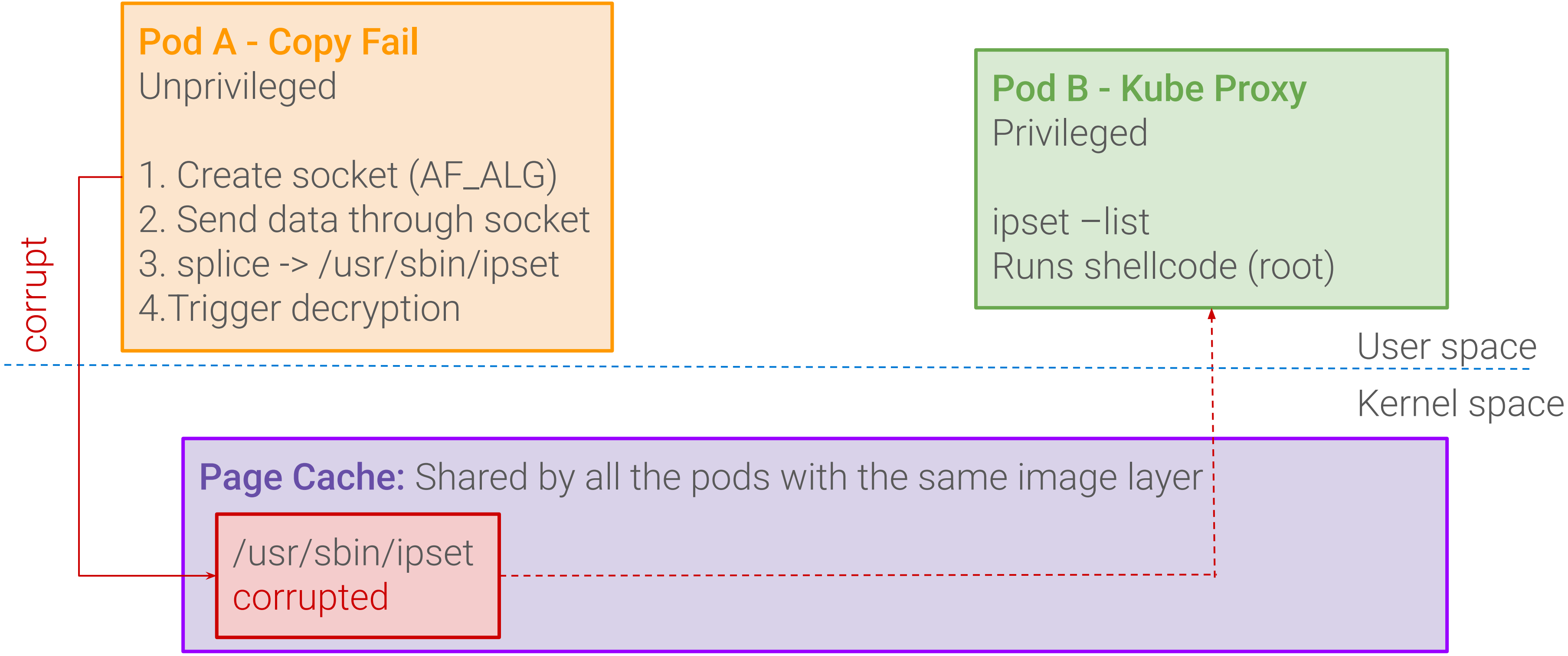
CROSS CONTAINER POISONING
Potentially backdoor co-located pods.

CONTAINER ESCAPE
A fully unprivileged container can achieve node-level execution.

Image Layer Sharing



Cross Container Propagation



The payload

```c
#include "nolibc/nolibc.h"

 int main(void) {
    const char msg[] = "CONTAINER ESCAPE SUCCESSFUL!\n";
    int fd;

    mkdir("/mnt", 0755);
    mount("/dev/sda1", "/mnt", "ext4", 0, NULL);

    fd = open("/mnt/root/res", O_WRONLY | O_CREAT | O_TRUNC, 0644);
    write(fd, msg, sizeof(msg) - 1);
    close(fd);

    umount2("/mnt", 2);
    return 0;

}
```

This is a cross-platform shellcode, built against the kernel's nolibc.

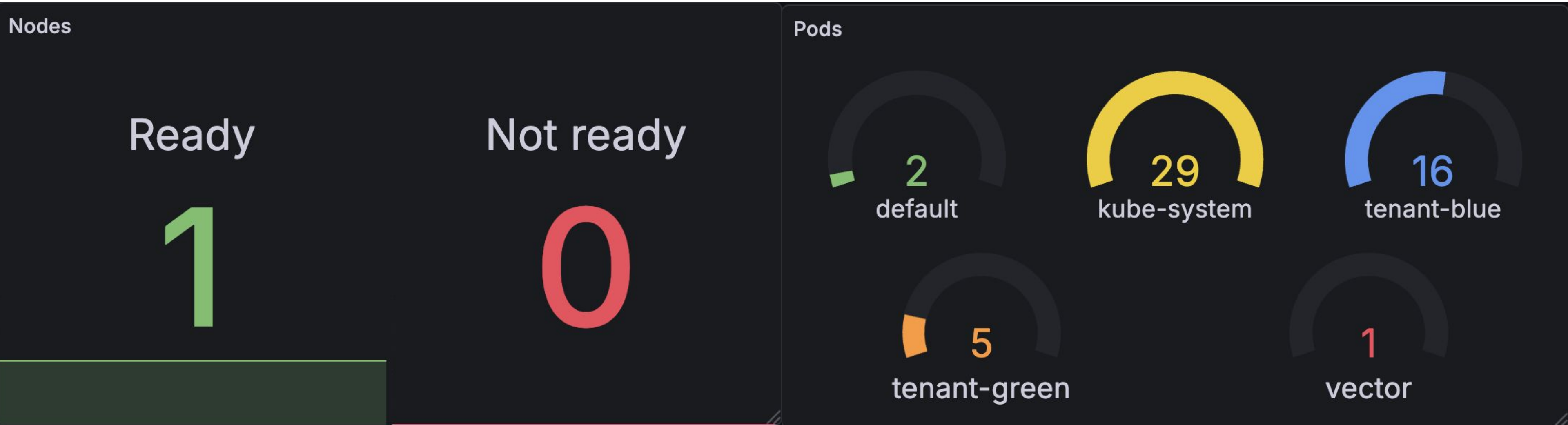
It mounts the node's filesystem, then creates and writes in /root/res to demonstrate access. Finally, it unmounts the filesystem not to leave persistent traces.

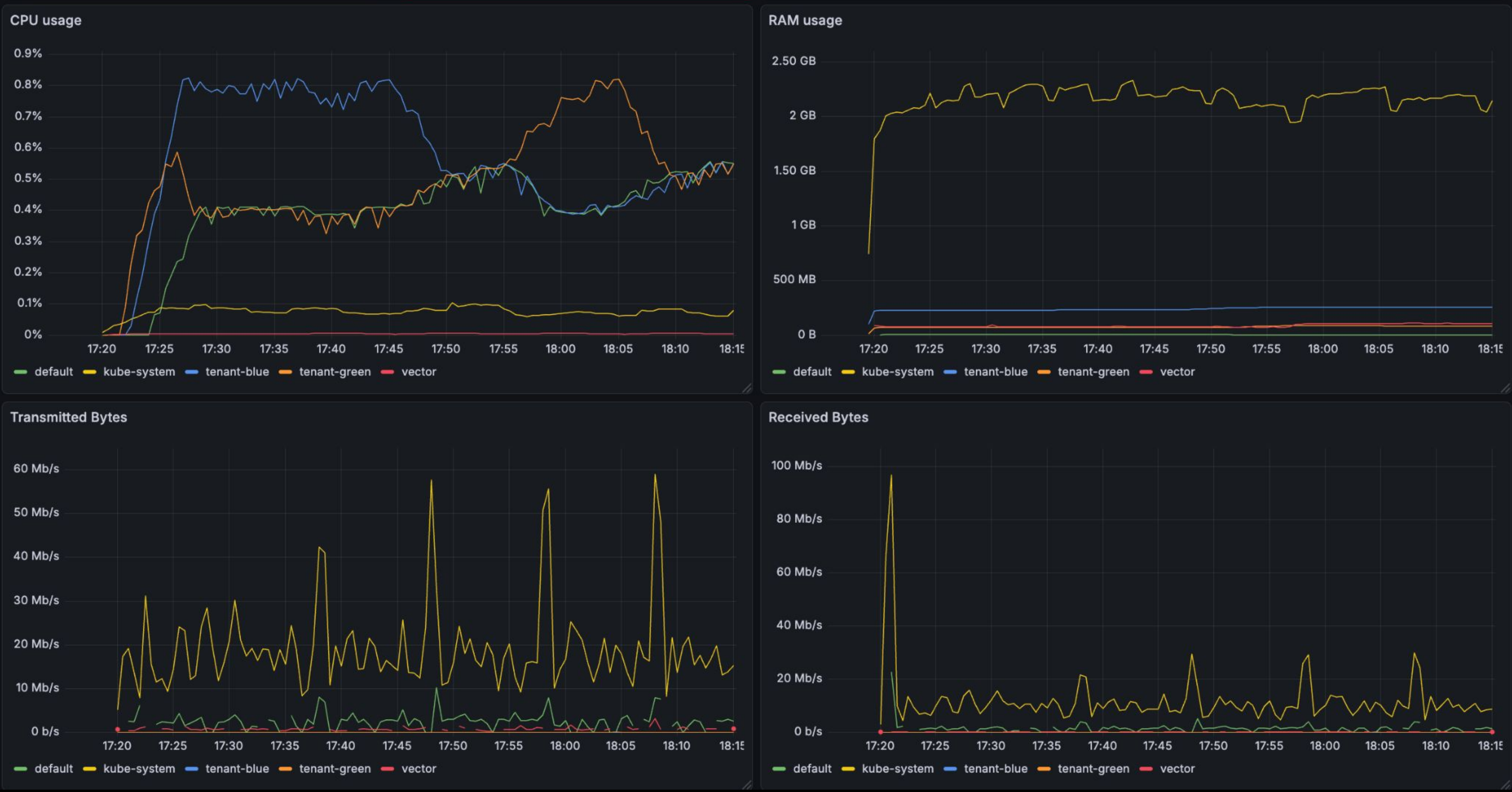
NOTE: the payload binary must be smaller than the ipset one to make the exploit work! Remember to add -O3 when compiling.

Azure Monitor

Azure Monitor is Microsoft's unified observability service for collecting, analyzing, and acting on telemetry from cloud and hybrid environments.

Using these metrics, we can build a Grafana dashboard.




CPU usage
0.9%
0.8%
0.7%
0.6%
0.5%
0.4%
0.3%
0.2%
0.1%
0%
17:20 17:25 17:30 17:35 17:40 17:45 17:50 17:55 18:00 18:05 18:10
default kube-system tenant-blue tenant-green vector
RAM usage
2.50 GB
2 GB
1.50 GB
1 GB
500 MB
0 B
17:20 17:25 17:30 17:35 17:40 17:45 17:50 17:55 18:00 18:05 18:10
default kube-system tenant-blue tenant-green vector
Transmitted Bytes
60 Mb/s
50 Mb/s
40 Mb/s
30 Mb/s
20 Mb/s
10 Mb/s
0 b/s
17:20 17:25 17:30 17:35 17:40 17:45 17:50 17:55 18:00 18:05 18:10
default kube-system tenant-blue tenant-green vector
Received Bytes
100 Mb/s
80 Mb/s
60 Mb/s
40 Mb/s
20 Mb/s
0 b/s
17:20 17:25 17:30 17:35 17:40 17:45 17:50 17:55 18:00 18:05 18:10
default kube-system tenant-blue tenant-green vector

cluster-forensics | Monitor

Kubernetes service

konnectivity-agent-6bc8c45497-ts9lj	Ok	-	-	-	13 mins	konnectivity-agent-6bc8c45497
konnectivity-agent-6bc8c45497-6g7ts	Ok	-	-	-	13 mins	konnectivity-agent-6bc8c45497
konnectivity-agent-autoscaler-5f7888c7	Ok	-	-	-	14 mins	konnectivity-agent-autoscaler-5f7888c78f
hubble-relay-7769d7d795-xd4jq	Ok	-	-	-	13 mins	hubble-relay-7769d7d795
csi-azurefile-node-qrqcx	Ok	-	-	-	14 mins	csi-azurefile-node
csi-azuredisk-node-flqsc	Ok	-	-	-	14 mins	csi-azuredisk-node
coredns-74f87d9649-xtdvf	Ok	-	-	-	14 mins	coredns-74f87d9649
coredns-74f87d9649-jfk52	Ok	-	-	-	14 mins	coredns-74f87d9649
coredns-autoscaler-5bdf6c6797-fljsq	Ok	-	-	-	14 mins	coredns-autoscaler-5bdf6c6797
copy-fail-poc-76c59fb745-xkm2s	Ok	-	-	-	2 mins	copy-fail-poc-76c59fb745
cloud-node-manager-bm9kd	Ok	-	-	-	14 mins	cloud-node-manager
azure-ip-masq-agent-bkc57	Ok	-	-	-	14 mins	azure-ip-masq-agent
azure-cns-ttmzv	Ok	-	-	-	14 mins	azure-cns
ama-metrics-operator-targets-fd9f769c	Ok	-	-	-	13 mins	ama-metrics-operator-targets-fd9f769c5
ama-metrics-node-jmxx4	Ok	-	-	-	13 mins	ama-metrics-node
ama-metrics-ksm-5b746c9b6f-2btmf	Ok	-	-	-	13 mins	ama-metrics-ksm-5b746c9b6f
ama-metrics-745f647957-z29pg	Ok	-	-	-	13 mins	ama-metrics-745f647957
ama-metrics-745f647957-f87tn	Ok	-	-	-	13 mins	ama-metrics-745f647957

copy-fail-poc | Overview

Container

View Logs in Log Analytics

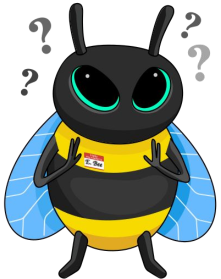
Overview   Live Logs   Live Events

Container Name	Container ID
copy-fail-poc	90220601c3e391052619ca52d6f06f8adfd649e191b205707d30c3fd59
Namespace	Container Status
tenant-blue	running
Container Status Reason	Image
-	docker.io/iochia02/copy-fail
Image Tag	Pod Creation Time Stamp
v0.19	14/05/2026, 17:29:15
Start Time	Finish Time
14/05/2026, 17:29:17	-
CPU Limit	CPU Request
Limit not set	Request not set
Memory Limit	Memory Request
Limit not set	Request not set
Last reported	
14 secs ago	

Live events

```
35 mins ago [Pod] [copy-fail-poc-76c59fb745-xkm2s] Scheduled: Successfully assigned tenant-blue/copy-fail-poc-76c59fb745-xkm2s to aks-agentpool-32433039-vmss00000g
35 mins ago [Pod] [copy-fail-poc-76c59fb745-xkm2s] Pulling: Pulling image "iochia02/copy-fail:v0.19"
35 mins ago [Pod] [copy-fail-poc-76c59fb745-xkm2s] Pulled: Successfully pulled image "iochia02/copy-fail:v0.19" in 725ms (725ms including waiting). Image size: 73828798 bytes.
35 mins ago [Pod] [copy-fail-poc-76c59fb745-xkm2s] Created: Created container: copy-fail-poc
35 mins ago [Pod] [copy-fail-poc-76c59fb745-xkm2s] Started: Started container copy-fail-poc
1 min ago [Pod] [copy-fail-poc-76c59fb745-xkm2s] Killing: Stopping container copy-fail-poc
```

What happened in this timeframe?

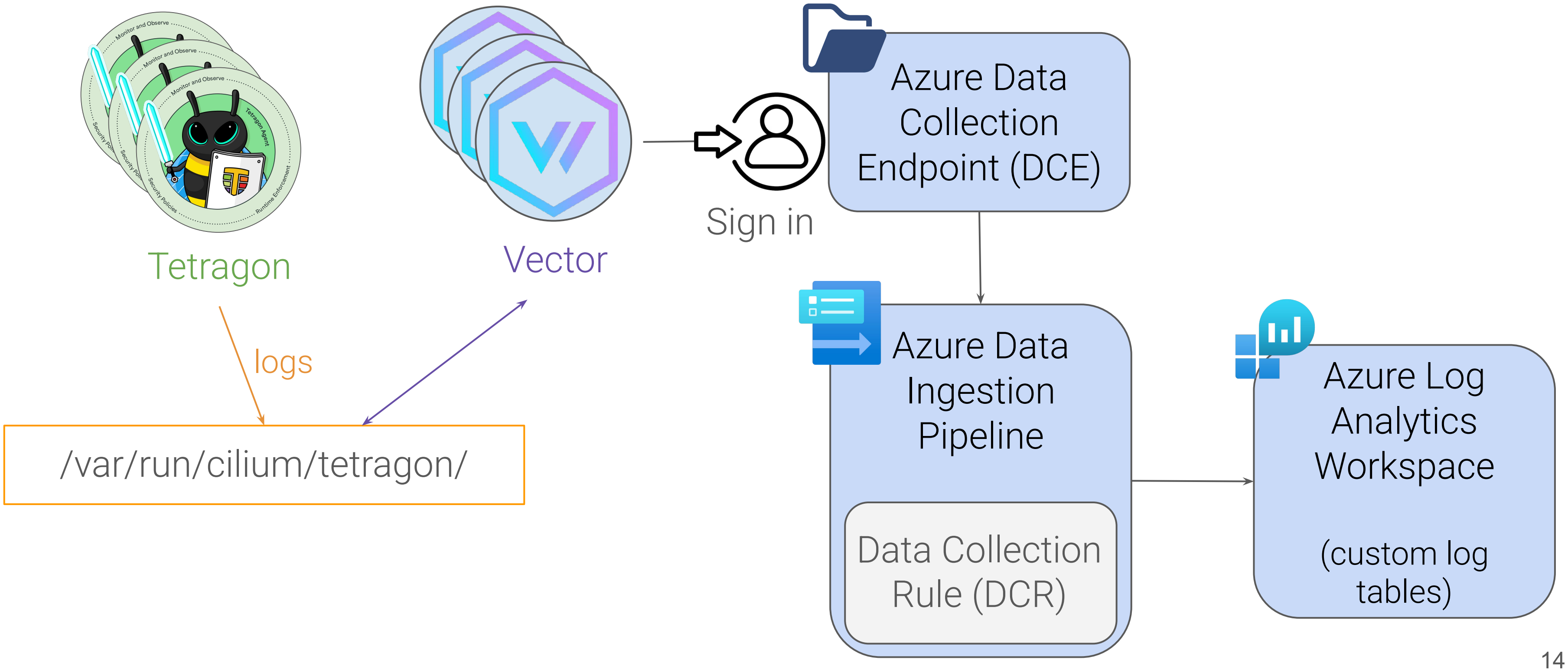


eBPF and Tetragon

eBPF is technology that enables event-driven custom code to run natively in an operating system kernel.

Tetragon is an open-source Kubernetes-aware security observability and runtime enforcement tool that applies policy and filtering directly with eBPF, allowing for reduced observation overhead, tracking of any process, and real-time enforcement of policies.

The monitoring pipeline



Tetragon Logs

Run the exploit

```
🚀 process tenant-blue/copy-fail-poc-76c59fb745-xkm2s /bin/copyfail --target /usr/sbin/ipset
💥 exit    tenant-blue/copy-fail-poc-76c59fb745-xkm2s /bin/copyfail --target /usr/sbin/ipset 0
🚀 process kube-system/kube-proxy-7t7bm /usr/sbin/iptables -w 5 -S KUBE-PROXY-CANARY -t mangle
💥 exit    kube-system/kube-proxy-7t7bm /usr/sbin/iptables -w 5 -S KUBE-PROXY-CANARY -t mangle 0
🚀 process kube-system/kube-proxy-7t7bm /usr/sbin/ip6tables -w 5 -S KUBE-PROXY-CANARY -t mangle
💥 exit    kube-system/kube-proxy-7t7bm /usr/sbin/ip6tables -w 5 -S KUBE-PROXY-CANARY -t mangle 0
🚀 process kube-system/kube-proxy-7t7bm /usr/sbin/iptables-save -t nat
💥 exit    kube-system/kube-proxy-7t7bm /usr/sbin/iptables-save -t nat 0
🚀 process kube-system/kube-proxy-7t7bm /usr/sbin/iptables-restore -w 5 --noflush --counters
💥 exit    kube-system/kube-proxy-7t7bm /usr/sbin/iptables-restore -w 5 --noflush --counters 0
🚀 process kube-system/kube-proxy-7t7bm /usr/sbin/ipset --version
💾 mount   kube-system/kube-proxy-7t7bm /usr/sbin/ipset /dev/sda1 /mnt
💥 exit    kube-system/kube-proxy-7t7bm /usr/sbin/ipset --version 0
```

The payload is executed

Note: this logs are accessible at execution time using the tetra cli.
However, thanks to our pipeline, they're stored in the LogAnalytics custom table, with a configurable retention time.

Log Analytics - Exploit

node_labels	{"agentpool":"agentpool","beta.kubernetes.io/arch":"amd64","beta.kubernetes.io/instance-type":"Standard_D2as_v4","beta.kubernetes.io/os":"linux","failure-domain.beta.kubernetes.io/region":"austriaeast"
node_name	aks-agentpool-32433039-vmss00000g
process_exec	{"parent":{"auid":4294967295,"binary":"/bin/bash","cwd":"/","docker":"ae3126fdc30e4054eab60cea7161be3","exec_id":"YWtzLWFnZW50cG9vbC0zMjQzMzAzOS12bXNzMDAwMDBnOjExMzQ5ODQv
parent	{"auid":4294967295,"binary":"/bin/bash","cwd":"/","docker":"ae3126fdc30e4054eab60cea7161be3","exec_id":"YWtzLWFnZW50cG9vbC0zMjQzMzAzOS12bXNzMDAwMDBnOjExMzQ5ODQwNjI2MDU6MzkxMTE=
process	{"arguments":"--target /usr/sbin/ipset","auid":4294967295,"binary":"/bin/copyfail","cwd":"/","docker":"ae3126fdc30e4054eab60cea7161be3","exec_id":"YWtzLWFnZW50cG9vbC0zMjQzMzAzOS12bXNzM
arguments	--target /usr/sbin/ipset
auid	4294967295
binary	/bin/copyfail
cwd	/
docker	ae3126fdc30e4054eab60cea7161be3
exec_id	YWtzLWFnZW50cG9vbC0zMjQzMzAzOS12bXNzMDAwMDBnOjExODYyMDgwNDQwNzY6Mzk4NzQ=
flags	execve rootcwd clone
in_init_tree	false
parent_exec_id	YWtzLWFnZW50cG9vbC0zMjQzMzAzOS12bXNzMDAwMDBnOjExMzQ5ODQwNjI2MDU6MzkxMTE=
pid	39874
pod	{"container":{"id":"containerd://ae3126fdc30e4054eab60cea7161be393f98a54a996df44d6b815ab60f53ca5b","image":{
start_time	2026-05-14T15:36:10.9954393Z
tid	39874
uid	0
TimeGenerated [UTC]	2026-05-14T15:36:11.767611Z
TenantId	a951cea8-e540-4234-ad41-694a7583f4ca
Type	CopyFail_CL

image	{"id":"docker.io/iochia02/copy-fail@sha
id	docker.io/iochia02/copy-fail@sha256:48e61
name	docker.io/iochia02/copy-fail:v0.19
name	copy-fail-poc
pid	21
security_context	{}
start_time	2026-05-14T15:29:17.0000000Z
name	copy-fail-poc-76c59fb745-xkm2s
namespace	tenant-blue
pod_labels	{"app.kubernetes.io/name":"copy-fail-poc","app
uid	db6eb1d8-fa14-48e9-813a-1b8a22fce03e
workload	copy-fail-poc

Log Analytics - Filesystem mount

node_labels	{"agentpool":"agentpool","beta.kubernetes.io/arch":"amd64","beta.kubernetes.io/instance-type":"Standard_D2as_v4","beta.kubernetes.io/os":"linux","failure-domain.beta.kubernetes.io/region":"austriaeast","failure-domain.be
node_name	aks-agentpool-32433039-vmss00000g
TimeGenerated [UTC]	2026-05-14T15:36:29.767479Z
process_kprobe	{"action":"KPROBE_ACTION_POST","args":[{"string_arg":"/dev/sda1"},{"string_arg":"/mnt"},{"string_arg":"ext4"}],"function_name":"__x64_sys_mount","parent":{"auid":4294967295,"binary":"/bin/bash","cwd":"/","docker
action	KPROBE_ACTION_POST
args	[{"string_arg":"/dev/sda1"},{"string_arg":"/mnt"},{"string_arg":"ext4"}]
function_name	__x64_sys_mount
parent	{"auid":4294967295,"binary":"/bin/bash","cwd":"/","docker":"c7a9bb90c86f0e1ba7abbe66a25a06c","exec_id":"YWtzLWFnZW50cG9vbC0zMjQzMzAzOS12bXNzMDAwMDBnOjExNjU4MjA1NTgyODY6Mzk1ODE=","flags":"exec
policy_name	monitor-mounts
process	{"arguments":"--version","auid":4294967295,"binary":"/usr/sbin/ipset","cwd":"/","docker":"c7a9bb90c86f0e1ba7abbe66a25a06c","exec_id":"YWtzLWFnZW50cG9vbC0zMjQzMzAzOS12bXNzMDAwMDBnOjEyMDM3M
arguments	--version
auid	4294967295
binary	/usr/sbin/ipset
cwd	/
docker	c7a9bb90c86f0e1ba7abbe66a25a06c
exec_id	YWtzLWFnZW50cG9vbC0zMjQzMzAzOS12bXNzMDAwMDBnOjEyMDM3MzM2MzQ5MzU6NDAzMDg=
flags	execve rootcwd clone
in_init_tree	false
parent_exec_id	YWtzLWFnZW50cG9vbC0zMjQzMzAzOS12bXNzMDAwMDBnOjExNjU4MjA1NTgyODY6Mzk1ODE=
pid	40308
pod	{"container":{"id":"containerd://c7a9bb90c86f0e1ba7abbe66a25a06cb5c38bbd07d2e8218549e668ee4af5057","im
refcnt	1
start_time	2026-05-14T15:36:28.5210300Z

container	{"id":"containerd://c7a9bb90c86f0e1ba7abbe66a25a06cb5c38bbd07d2
name	kube-proxy-7t7bm
namespace	kube-system
pod_labels	{"component":"kube-proxy","controller-revision-hash":"746f4bfd65","ku
uid	a5904d2a-633e-4b1c-a041-9574c91cffb5
workload	kube-proxy
workload_kind	DaemonSet

Detect vulnerability exploit

When the vulnerability becomes publicly known, we can write a policy to detect the usage of the AF_ALG socket.

```
🚀 process tenant-blue/copy-fail-poc-76c59fb745-xkm2s /bin/copyfail --target /usr/sbin/ipset
❓ syscall tenant-blue/copy-fail-poc-76c59fb745-xkm2s /bin/copyfail security_socket_bind
❓ syscall tenant-blue/copy-fail-poc-76c59fb745-xkm2s /bin/copyfail security_socket_bind
❓ syscall tenant-blue/copy-fail-poc-76c59fb745-xkm2s /bin/copyfail security_socket_bind
❓ syscall tenant-blue/copy-fail-poc-76c59fb745-xkm2s /bin/copyfail security_socket_bind
❓ syscall tenant-blue/copy-fail-poc-76c59fb745-xkm2s /bin/copyfail security_socket_bind
💥 exit    tenant-blue/copy-fail-poc-76c59fb745-xkm2s /bin/copyfail --target /usr/sbin/ipset 0
🚀 process kube-system/kube-proxy-7t7bm /usr/sbin/ipset --version
💾 mount   kube-system/kube-proxy-7t7bm /usr/sbin/ipset /dev/sda1 /mnt
💥 exit    kube-system/kube-proxy-7t7bm /usr/sbin/ipset --version 0
🚀 process kube-system/kube-proxy-7t7bm /usr/sbin/iptables -w 5 -S KUBE-PROXY-CANARY -t mangle
💥 exit    kube-system/kube-proxy-7t7bm /usr/sbin/iptables -w 5 -S KUBE-PROXY-CANARY -t mangle 0
🚀 process kube-system/kube-proxy-7t7bm /usr/sbin/ip6tables -w 5 -S KUBE-PROXY-CANARY -t mangle
💥 exit    kube-system/kube-proxy-7t7bm /usr/sbin/ip6tables -w 5 -S KUBE-PROXY-CANARY -t mangle 0
```

Block vulnerability exploit

But using Tetragon we can even block the socket creation to prevent the exploit!

```
🚀 process tenant-blue/copy-fail-poc-76c59fb745-xkm2s /bin/copyfail --target /usr/sbin/ipset
❓ syscall tenant-blue/copy-fail-poc-76c59fb745-xkm2s /bin/copyfail security_socket_bind
💥 exit    tenant-blue/copy-fail-poc-76c59fb745-xkm2s /bin/copyfail --target /usr/sbin/ipset SIGKILL
🚀 process kube-system/kube-proxy-7t7bm /usr/sbin/ipset --version
💥 exit    kube-system/kube-proxy-7t7bm /usr/sbin/ipset --version 0
```

node_labels	{"agentpool":"agentpool","beta.kubernetes.io/arch":"amd64","beta.kubernetes.io/instance-type":"Standard_D2as_v4","beta.kubernetes.io/os":"linux","failure-domain.beta.kubernetes.io/region":"austriaeast","failure-
node_name	aks-agentpool-32433039-vmss00000g
TimeGenerated [UTC]	2026-05-14T15:51:50.767451Z
process_kprobe	{"action":"KPROBE_ACTION_SIGKILL","args":[{"sockaddr_arg":{"addr":"invalid IP","family":"AF_ALG"}}],"function_name":"security_socket_bind","parent":{"auid":4294967295,"binary":"/bin/bash","cwd":"/","do
action	KPROBE_ACTION_SIGKILL
args	[{"sockaddr_arg":{"addr":"invalid IP","family":"AF_ALG"}}]
function_name	security_socket_bind
parent	{"auid":4294967295,"binary":"/bin/bash","cwd":"/","docker":"ae3126fdc30e4054eab60cea7161be3","exec_id":"YWtzLWFnZW50cG9vbC0zMjQzMzAzOS12bXNzMDAwMDBnOjExMzQ5ODQwNjI2MDU6MzkxMTE=",
policy_name	alg-bind2
process	{"arguments":"--target /usr/sbin/ipset","auid":4294967295,"binary":"/bin/copyfail","cwd":"/","docker":"ae3126fdc30e4054eab60cea7161be3","exec_id":"YWtzLWFnZW50cG9vbC0zMjQzMzAzOS12bXNzMDA
arguments	--target /usr/sbin/ipset
auid	4294967295
binary	/bin/copyfail
cwd	/

Final forensic considerations

1. Kernel-Level Security

Tetragon captures events in Kernel Space at system calls invocation, preventing TOCTOU attacks. Attackers cannot modify hooks or logs from unprivileged containers.

2. Enriched Forensic Context

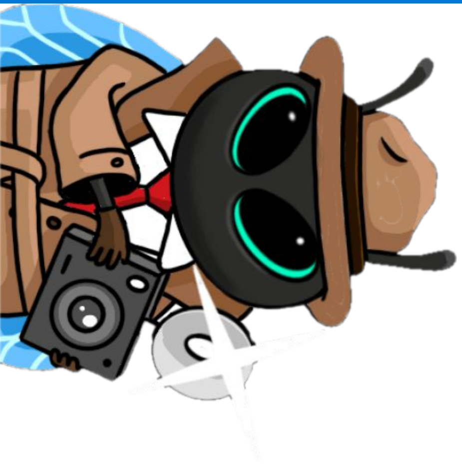
Tetragon enriches logs with kubernetes-aware information (container IDs, pod namespaces), processes information (PIDs, parent process) and users information.

3. Real-time streaming

Vector immediately sends the logs to Log Analytics, reducing the WoE. It can be configured not to be able to modify/delete already pushed logs.

4. Pipeline Integrity

Vector uses TLS 1.2/1.3 to securely stream payloads, blocking a MitM. Entra ID authN prevents unauthorized actors from injecting fake logs.



Thank you!
Questions?